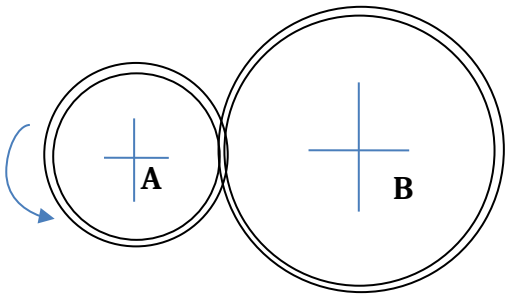
Simple and Compound Gear Train Worksheet

Q1.

A simple gear train is shown below.



A = 15 teeth

B = 45 teeth

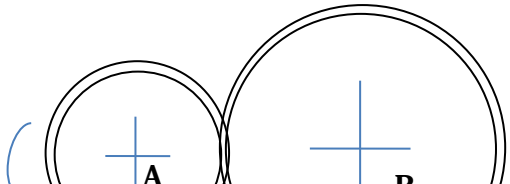
If the driver, A, rotates at 900 revs min^{-1}, calculate:

(a) The velocity ratio.

(b) The speed of the driven gear.

Q2.

A simple gear train is used in a conveyor system.



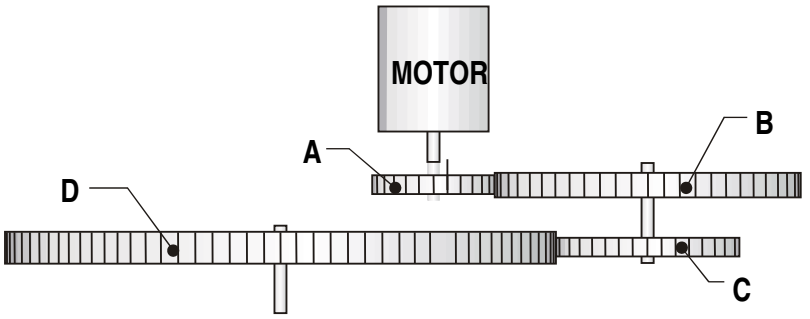
The driver gear, A, has 20 teeth and the driven gear, B, has 60 teeth.

(a) Calculate the velocity ratio.

(b) If the input speed is 720 revs min^{-1}, determine the output speed.

Q3.

A compound gear train is shown below.



A = 25 teeth
B = 75 teeth
C = 20 teeth
D = 80 teeth

The motor drives gear A at 1000 revs min^{-1}.

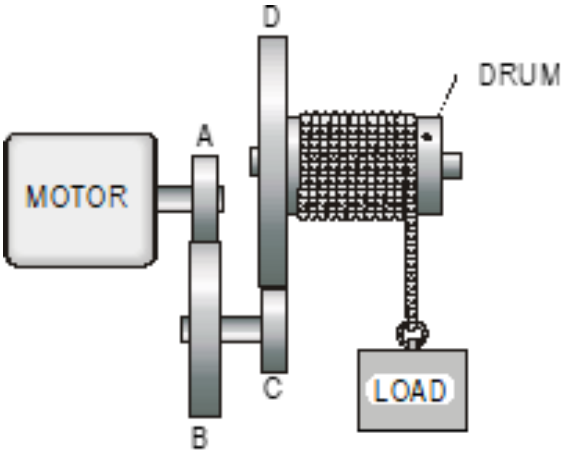
(a) Calculate the velocity ratio of the system.

Substitute both velocity ratios into the original formula,

(b) Determine the output speed of gear D.

Q4.

A winch mechanism uses a compound gear train.



A = 12 teeth
B = 48 teeth
C = 18 teeth
D = 54 teeth

If the drum attached to D rotates at 50 revs min^{-1}, calculate:
(a) The overall velocity ratio.

Substitute both velocity ratios into the original formula,

(b) The motor speed required to achieve this output.

Q5.

A simple gear train has a driver gear of 10 teeth and a driven gear of 40 teeth.

(a) What is the velocity ratio?

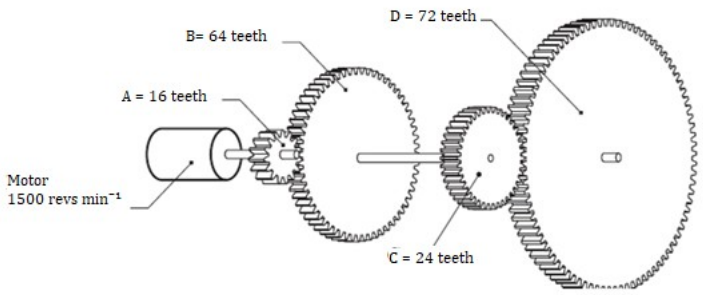
(b) If the driver rotates at 600 revs min^{-1}, what is the output speed?

(c) State the direction of rotation of the output gear compared to the driver.

Output gear (driven) will rotate in the opposite direction to driver.

Q6.

A compound gear system for a small lathe is shown.



A = 16 teeth
B = 64 teeth
C = 24 teeth
D = 72 teeth

The motor drives gear A at 1500 revs min^{-1}.

(a) Calculate the velocity ratio.

Substitute both velocity ratios into the original formula,

(b) Determine the speed of the output shaft.

Q7.

A simple gear train is designed to increase speed.

A = 30 teeth
B = 10 teeth

If A rotates at 300 revs min^{-1}, calculate:
(a) The velocity ratio.

Remember, the smaller gear will always rotate faster, therefore Gear B must be the output or driven gear if the system is to increase speed.

(b) The speed of gear B.

Q8.

A compound gear train has the following data:

A = 18 teeth
B = 72 teeth
C = 24 teeth

D = 60 teeth

The input rotates at 1200 revs min^{-1}.

(a) Find the velocity ratio.

Substitute both velocity ratios into the original formula,

(b) Find the speed of the final output gear.

Q9.

A simple gear train is used in a model vehicle.

The driver has 12 teeth, the driven has 36 teeth.

(a) Determine the velocity ratio.

(b) If the driver runs at 1800 revs min^{-1}, calculate the speed of the driven gear.

(c) State one reason for using plastic gears in this model.

Plastic gears are low cost compared to metal gears, they are lighter in mass than metal gears and have an element of self-lubricity (reducing friction and in turn heat)

Q10.

A compound gear train for a hoist mechanism has the following gears:

A = 14 teeth
B = 70 teeth
C = 20 teeth
D = 80 teeth

If the hoist drum (gear D) turns at 20 revs min^{-1}, calculate:

(a) The required motor speed.

Substitute both velocity ratios into the original formula,

(b) The overall velocity ratio.

From (a),